# IC improvement round 2

# IC improvement round 2: results (2026-10-15)

These are follow-ups to `../ic_improve_20261014/README.md`, run under `RULES.md`:

- Every candidate is compared with the incumbent IC component it replaces.
- Every comparison is paired, uses the same counted unit on both arms, and verifies every logarithm or root set.
- The candidates were tuned on training data, frozen, then tested on held-out data.

| # | Candidate vs incumbent | Verdict | Level | Key numbers | Landed |
|---|---|---|---|---|---|
| 02 | Free relations in the production compact-orbit Koblitz IC, K tuned per arm | **ADVANCES** | constant | | **Default on** in `examples/koblitz_orbit_dlp_fast_online.rs`. `KIC_FREE_RELATIONS=0` restores the old pipeline byte-identically (A/A 24/24). |
| 04 | k = 4 IC: first-dependency stop (d4-ed) vs $D_4$. This fixes round 1's $D_4$+2LP regression. | **ADVANCES** | constant (exponent MATCHES) | | ecbench base `yline-d4-ed`. Existing hashes are unchanged. |
| 05 | Koblitz Gröbner oracle: harvest the multi-variable linear rows that `decisive_rows` discards, root only, gated on generators per variable < 0.9607 | **ADVANCES** where the gate fires; MATCHES elsewhere | constant | | **Default on** in `koblitz_groebner.rs`. `KIC_LINEAR_HARVEST=0` restores the old engine (counters 60/60). |
| 03 | Koblitz Gröbner oracle: gated fall-row augmentation | ADVANCES where it fires; production no-op | constant | | Opt-in `KIC_FALLROW_GATE=1`. |
| 01 | Genus-3 2LP IC: large primes as columns + structured Gaussian elimination + Wiedemann (`lpsgewd`) vs round-1 IC-2LP (DFS substitution + Lanczos on normal equations) | **ADVANCES** | constant; exponent small | | example only |

## Open follow-ups

- **Koblitz K per size.** Retuning K alone at n = 41 (255 → 100) gives 0.46×, with no new code. The recommended values are K = 125 / 280 / 500 for n = 41 / 53 / 61. Item 02 left the callers' K unchanged; that is an owner decision.
- **Linear-harvest gate.** Generators per variable is close to (N−1)/N on pipeline systems, so the gate skips the largest gain: ungated root harvest gives 1.28× at (29,15) m2. A gate on variable count or tree size needs its own protocol.
- **Linear harvest end to end at (23,12).** The bench gain (1.17×) does not carry through: the production path stops at the first accepted root, so its solves are shorter than the bench's.
- **Genus-3 linear algebra.** Elimination removes a shrinking share of columns: it keeps 32% of the incumbent's columns at 2^10 but 91% at 2^20. Weight growth turns into column growth, so linear algebra stays near q^1.54 locally. The next levers are a κ retune at large q (exploratory κ = 1.41 is the cheapest arm at 2^21) and block Wiedemann.

## Caveats

- **Host:** shared, 4 vCPU, load average 3–13, isolation level L0. Verdicts use counted operations or thread CPU.
- **Scale:** sizes are small.
- **Conductor:** the control plane was unreachable.